

Problem statement and Q&A

Focus Areas

- Develop AI/ML-based feature extraction models capable of identifying key structures (buildings, roads, water bodies, utilities, etc.) from high-resolution drone orthophotos.
- Create automated DTM and drainage network modeling workflows from point-cloud datasets, improving efficiency in hydrological design and land management.
- Promote innovation and research collaboration among academia, industry, and government to strengthen India's geospatial intelligence ecosystem.
- Enhance the accuracy, scalability, and reliability of geospatial data processing for rural mapping and planning applications.
- Encourage participants to develop open, interoperable, and scalable geospatial solutions that can be adapted across diverse Indian terrains.

1. Problem Statement 1 : Feature Extraction from Drone Images:

i. Develop an AI model capable of identifying key features in orthophotos with high precision. Use of AI/ML techniques for extraction of the following features from SVAMITVA Drone Imagery: -

- Building footprint extraction (built-up area from the drone image and classified roof-top based on observation of the imagery as RCC, Tiled, Tin, and Others.
- Road feature extraction
- Waterbodies extraction,
- Distribution Transformer location/ Over-head Tank, well location identification etc.

ii. Achieve a target accuracy of 95% in feature identification.

iii. Optimize the model for efficient processing and deployment.

Input Data

- Drone imagery for 10 villages (training & validation).
- Drone imagery for 10 additional villages (output testing).

Expected Deliverables

SVAMITVA Scheme drone-images for 10 villages along with feature extracted datasets for 10 villages to train and validate the AI model, and 10 more villages drone data for output testing.

- A fully trained and optimized AI model for feature identification in orthophotos.
- Documentation detailing the model architecture, training process, and deployment guidelines.
- A final report summarizing the project outcomes, including accuracy metrics and recommendations for future improvements.

2. Problem Statement 2: DTM Creation using AI/ML from point cloud data and development of drainage network

To conceptualize and develop a data-driven, DTM using Drone point cloud datasets leveraging AI/ML:

- Delineate natural surface-water flow paths and low-lying zones,
- Predict waterlogging hotspots, and
- Design an optimized, resilient drainage network for densely inhabited village (abadi) areas.

Input data:

- Point Cloud data for 10 villages

Expected Deliverables

- Automated AI/ML Processing: From point-cloud classification to generate DTM.
- Optimized Drainage Network Design: GIS-ready layers and design parameters.
- Documentation detailing the model architecture, training process, and deployment guidelines.
- A final report summarizing the project outcomes, including accuracy metrics and recommendations for future improvements.

Expected Outcomes

- Development of working prototypes or proof-of-concept models integrating AI/ML, Drones, and GIS.
- Generation of AI-driven digital maps, DTMs, and decision-support systems for real-world applications.
- Creation of an innovation ecosystem connecting academic research with industry needs. Strengthening the foundation for digital governance and data-driven infrastructure planning.
- Identification of talent and technology solutions for potential opportunities.

Frequently Asked Questions

1. **What does “Idea Submission” mean during registration? Ideally, this is relevant during submission ,not at the time of registration.**

Not Mandatory. At the time of registration, participants are required to submit a brief concept note outlining their understanding of the problem, proposed approach, and intended methodology. This is not the final solution, but an expression of intent and technical direction.

The detailed implementation, models, and outputs are expected during the final submission phase of the hackathon.

2. **When and how will access to the dataset be provided?**

Datasets will be shared after the registration closes, as per the timeline, during the dataset-sharing phase.

Access details (download links / credentials) will be communicated only to registered and eligible teams via email.

3. **Are participants expected to create ground truth labels, or will IIT-T provide training, validation, and test splits?**

Ground truth data and appropriate training/validation/test splits will be provided as part of the dataset to ensure consistency and fair evaluation. We will be sharing required details subsequently.

Participants are expected to focus on model development, optimisation, and innovation, not on manual ground-truth creation.

4. **Will you provide COMPUTE platform (CPU/GPU), or is it the participant’s responsibility?**

Participants are expected to use their own compute resources (CPU/GPU/cloud).

The hackathon is designed to remain platform-agnostic, encouraging scalable and deployable solutions. Any specific guidance, if applicable, will be communicated separately.

5. **What is the required output format – GeoJSON or Shapefile?**

Participants may submit outputs in standard GIS-compatible formats, such as: GeoJSON, Shapefile

With the support of OGC India, we might include

- Raster outputs (orthophotos, DTM, derived surfaces) may be provided in Cloud Optimized GeoTIFF (COG) format.

- Vector outputs (building footprints, roads, drainage networks, asset locations, etc.) may be provided in GeoPackage (GPKG) format.

The final accepted format will be specified clearly in the submission guidelines shared during the hackathon phase.

6. How should models, results, and outputs be submitted?

Final submissions will include:

- Model outputs and extracted layers
- A brief technical report (methodology, accuracy, limitations)
- Supporting documentation

Submission will be through a designated submission portal / shared drive, details of which will be communicated to participants closer to the submission deadline.

7. Who should be contacted for quick clarification (WhatsApp, mobile, or email)?

For all official communication and clarifications, participants should contact:

 geointel.mopr@iittnif.com